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Roll No :

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Assignment no :

2

Submitted to :

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Subject :

Operational Research

Given Problem:

Maximize $Z = 6x_1 + 4x_2$

Subject to:

- $x_1 + 2x_2 \leq 720$
- $2x_1 + x_2 \leq 780$
- $x_1 \leq 320$
- $x_1, x_2 \geq 0$

Step 1:

Convert to Standard form.

Add Slack variables $s_1, s_2, s_3 \geq 0$:

- $x_1 + 2x_2 + s_1 = 720$
- $2x_1 + x_2 + s_2 = 780$
- $x_1 + s_3 = 320$

Objective Function:

Maximize $Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3$

OR in standard form: $Z - 6x_1 - 4x_2 = 0$

DATE: / /

Step 2:

initial Simplex table.

Basic ✓	x_1	x_2	s_1	s_2	s_3	Sol	Ratio
Z	-6	-4	0	0	0	0	7
s_1	1	2	1	0	0	720	$720/1 = 720$
s_2	2	1	0	1	0	780	$780/2 = 390$
s_3	1	0	0	0	1	320	$320/1 = 320$
Key row							

⇒ Most negative coefficient in Z-row: -6
(x_1 column) Pivot column: x_1 Minimum
ratio: 320 & Pivot element: 1 at (s_3, x_1)

$$s_2 \leftarrow 2x_1, s_1 \leftarrow x_1, Z \leftarrow 6x_1$$

By row operation

$$\begin{array}{l} s_2: 2 \quad 1 \quad 0 \quad 1 \quad 0 \quad 780 \\ 2x_1: 2 \quad 0 \quad 0 \quad 0 \quad 2 \quad 640 \\ \hline 0 \quad 1 \quad 0 \quad 1 \quad -2 \quad 140 \end{array}$$

$$s_1: 1 \quad 2 \quad 1 \quad 0 \quad 0 \quad 720$$

$$\begin{array}{l} x_1: 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 320 \\ \hline 0 \quad 2 \quad 1 \quad 0 \quad -1 \quad 400 \end{array}$$

$$Z: -6 \quad -4 \quad 0 \quad 0 \quad 0 \quad 0$$

$$\begin{array}{l} 6x_1: 6 \quad 0 \quad 0 \quad 0 \quad 6 \quad 1920 \\ \hline 0 \quad -4 \quad 0 \quad 0 \quad 6 \quad 1920 \end{array}$$

DATE: / /

table after iteration 1:

Basic ↓	x_1	x_2	key column			Sol	Ratio
Z	0	-4	0	0	6	1920	
S1	0	2	1	0	-1	400	$400/2=200$
S2	0	1	0	1	-2	140	$140/1=140$
key row X1	1	0	0	0	1	320	-

→ most negative coefficient: -4 (x_2 column)

Pivot Column: x_2 Minimum Ratio: 140 (S_2 row)

Pivot element: 1 at (S_2, x_2)

By row operation: ($S_1 - 2x_2, Z + 4x_2$)

$$S1: \quad 0 \quad 2 \quad 1 \quad 0 \quad -1 \quad 400$$

$$2x_2: \quad 0 \quad 2 \quad 0 \quad 2 \quad -4 \quad 280$$

$$\quad \quad 0 \quad 0 \quad 1 \quad -2 \quad 3 \quad 120$$

$$Z: \quad 0 \quad -4 \quad 0 \quad 0 \quad 6 \quad 1920$$

$$4x_2: \quad 0 \quad 4 \quad 0 \quad 4 \quad -8 \quad 560$$

$$\quad \quad 0 \quad 0 \quad 0 \quad 4 \quad -2 \quad 2480$$

Final table:

Basic ↓	x_1	x_2	s_1	s_2	s_3	Solution
Z	0	0	0	4	-2	2480
S1	0	0	1	-2	3	120
x_2	0	1	0	1	-2	140
x_1	1	0	0	0	1	320

⇒ All coefficients in Z-row are non-negative (except for slack variables which is acceptable).

DATE _/ _/ _

Solution ::

Optimal values:

• $x_1 = 320$

• $x_2 = 140$

Maximum value $z = 2400$

verification ::

• Constraint 1: $x_1 + 2x_2 \leq 720$

$$320 + 2(140) = 320 + 280 = 600 \leq 720 \checkmark$$

• Constraint 2: $2x_1 + x_2 \leq 780$

$$2(320) + 140 = 640 + 140 = 780 \leq 780 \checkmark$$

• Constraint 3: $x_1 \leq 320$

$$320 \leq 320 \checkmark$$

$$z = 6x_1 + 4x_2$$

$$= 6(320) + 4(140) = 1920 + 560 = 2480 \checkmark$$